

LESSON 3-4 • HOMEWORK PACKET • SHEET 1 OF 2**Graphing Ratios on the Coordinate Plane**

Maryland Standard 6.RP.A.3a • Learning target – I can turn a ratio into ordered pairs and plot them on a coordinate plane.

REMEMBER

- Write each pair from a ratio table as an **ordered pair** **(x, y)**. The first quantity is x.
- Points for equivalent ratios lie on a **straight line** through **(0, 0)**.

WORKED EXAMPLE

A pack holds 4 juice boxes. Let x = packs and y = juice boxes. Write ordered pairs for 1, 2, and 3 packs.

Step 1 Table: 1 pack, 4 boxes; 2 packs, 8 boxes; 3 packs, 12 boxes

Step 2 Ordered pairs: (1, 4), (2, 8), (3, 12)

Answer (0, 0), (1, 4), (2, 8), (3, 12) lie on one line.

PRACTICE

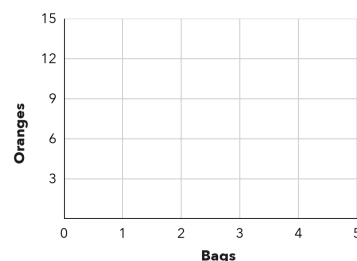
show your work • label units

1. Each bag holds 3 oranges. Complete the table. Then write each column as an ordered pair.

Bags (x)	1	2	3	4
Oranges (y)	3			
(x, y)	(1, 3)			

2. Plot the ordered pairs from Exercise 1. Draw a line through the points. Does the line pass through (0, 0)? What does (0, 0) mean?

(0, 0) means _____



3. A muffin batch uses 2 cups of flour (x = batches, y = cups). Write the ordered pair for:

a. 7 batches _____

b. 18 cups of flour _____

4. **Garden Rows.** Each row in a garden has 5 plants. Let x = rows and y = plants.

a. Write the ordered pair for 8 rows. _____

b. Write the ordered pair for 0 rows. What does it mean? _____

5. **Same Line?** Do the points (1, 5), (2, 10), and (3, 14) all lie on one straight line through (0, 0)?

☐ Yes

☐ No

6. **Find the Error.** Tickets cost \$2 each. On a graph with x = tickets and y = cost, Ruby plotted (12, 6) for 6 tickets. Explain her mistake and write the correct point.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____

AT HOME Make a table for something that comes in packs at home. Say two of its ordered pairs.

LESSON 3-4 • HOMEWORK PACKET • SHEET 2 OF 2**Interpreting Points on Ratio Graphs**

Maryland Standard 6.RP.A.3a • Learning target – I can explain what a point on a ratio graph means in the situation it describes.

REMEMBER

- Each point (x, y) is a pair of amounts. Say **both** with units.
- The point **$(1, y)$** shows the **unit rate**: $y \div x$ for any point. $(0, 0)$ means zero of both.

WORKED EXAMPLE

A graph of hours worked (x) and dollars earned (y) passes through $(3, 45)$. What does the point mean?

Step 1 $(3, 45)$: 3 hours of work earns \$45

Step 2 Unit rate: $45 \div 3 = 15$, so $(1, 15)$ is on the line

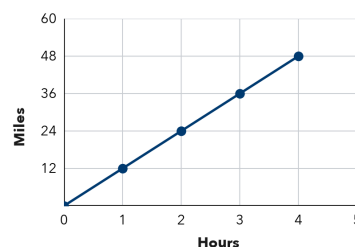
Answer \$15 per hour

PRACTICE

show your work • label units

1. The graph shows time (x, hours) and distance (y, miles) for a cyclist.

- What does the point $(2, 24)$ mean? _____
- What is the unit rate? _____ miles per hour
- At this rate, how far does the cyclist ride in 6 hours? _____ miles



2. A ratio graph of trays (x) and muffins (y) passes through $(1, 12)$. Find each missing value.

- $x = 4, y = \underline{\hspace{2cm}}$
- $y = 60, x = \underline{\hspace{2cm}}$
- $x = 7, y = \underline{\hspace{2cm}}$
- $y = 36, x = \underline{\hspace{2cm}}$
- $x = 10, y = \underline{\hspace{2cm}}$
- $y = 96, x = \underline{\hspace{2cm}}$

3. **Same Line?** A ratio graph of pounds of grapes (x) and cost in dollars (y) passes through $(4, 10)$.

- What is the unit rate? _____ per pound
- Is $(6, 15)$ on the line? Is $(8, 18)$? Show how you know. _____

4. **Find the Error.** A graph of laps (x) and minutes (y) passes through $(3, 12)$. Omar says it means 12 laps in 3 minutes. What is wrong?

TEST PRACTICE

choose the best answer

5. A ratio graph passes through $(5, 40)$. Which point is also on the line?

- A** $(1, 8)$ **B** $(6, 41)$ **C** $(8, 5)$ **D** $(10, 45)$

6. A graph of hours (x) and dollars earned (y) passes through $(2, 18)$. What does $(1, 9)$ mean?

- A** \$9 per hour **B** 9 hours for \$1 **C** \$18 in 1 hour **D** \$2 per hour

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Tell someone at home what $(1, 9)$ means on a graph of hours and dollars.